Problems, Programs, and Ideal Complexity

1 Problems

1.1 Definition of Problem $\mathcal{Q}$

A problem is also referred to as a "Question" - a bounded boolean expression

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow A_o \in \Omega \quad \forall X_n \in \mathbb{D}_f$$

1.2 True False Question

Also referred to as a "Decision Problem"

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_f$$

1.3 Numerical Problem

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow y_o \in \mathbb{C} \quad \forall X_n \in \mathbb{D}_f$$

1.4 System Question

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{Q} := f[X_n] \rightarrow A_o \in \Omega \quad \forall X_n \in \mathbb{D}_f :$$
$$|A_o| == M$$

2 Instructions

2.1 Definition of Instruction

An instruction is either a canonical comparison, canonical operation, or canonical measurement of Question $\mathcal{Q}$/ Framework $\mathcal{F}$/ System $\mathcal{S}$ followed by an assigment

2.2 Definition of Instruction Set

Define $\mathcal{I}$; an ordered set of instructions s_i

$$\mathcal{I} := \{s_1, s_2, ..., s_N\}$$

3 Memory

3.1 Definition of Memory Set

Define Memory $\mathcal{M}$; an ordered set of objects m_i

$$\mathcal{M} := \{m_1, m_2, ..., m_M\}$$

4 Programs

4.1 Definition of a Program

Define Program P

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \rightarrow y_o \in \Omega \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \,|y_o, b_2, b_3, ..., b_M\} \rightarrow y_o \in \Omega \quad \forall X_n \in \mathbb{D}_P$$

4.2 Void Programs

Define a void program; a program with inputs x_i and no output

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M\} \quad \forall X_n \in \mathbb{D}_P$$

4.3 Boolean Programs

Define a Boolean Program
Given

$$X_n = \{x_1, ..., x_n\}$$

$$\mathcal{P} := P[X_n] \rightarrow y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_P \quad =$$
$$\{s_1, s_2, ..., s_N \mid b_2, b_3, ..., b_M, y_o\} \rightarrow y_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_n \in \mathbb{D}_P$$

4.4 Numerical Programs

Define a numerical program; a program with inputs x_i, input set C, and real, rational output y_o

$$X_n = \{x_1, ..., x_n\}$$
$$\mathcal{P} := P[X_n] \rightarrow y_o \in \mathbb{Q} \wedge y_o \geqslant 0 \quad \forall X_n \in \mathbb{D}_P =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, y_o\} \rightarrow y_o \in \mathbb{Q} \wedge y_o \geqslant 0 \quad \forall X_n \in \mathbb{D}_P =$$

4.5 System Programs

Define a system program; a program with inputs x_i, input set C, and real, output set Y_o

$$X_n = \{x_1, ..., x_n\}$$
$$\mathcal{P} := P[X_n] \rightarrow Y_o \in \Omega = \{y_1, y_2, ..., y_K\} \in \Omega \quad \forall X_n \in \mathbb{D}_P \quad =$$
$$\{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, Y_o\} \rightarrow Y_o \in \Omega = \{y_1, y_2, ..., y_K\} \in \Omega \quad \forall X_n \in \mathbb{D}_P$$

5 Program Complexity

5.1 Definition of Computational Complexity of Serial Program $\mathcal{P}$

Define computational complexity of Program $\mathcal{P}$ with dimension C

$$\mathbf{O}[n] := \ < O_T[n], O_S[n], O_3[n], ..., O_C[n] >$$

5.2 Definition of Ideal Computational Complexity of Serial Program $\mathcal{P}$

Define ideal computational complexity of Program $\mathcal{P}$ with dimension 2

$$\mathbf{O}[n] := \ < O_T[n], O_S[n] >$$

5.3 Assumption 1

The remainder of this document assumes complexity means ideal computational complexity with dimension 2.

5.4 Total Complexity

Define Total Complexity of Boolean Program $\mathcal{P}$

$$O[n] := |P[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o| = N + M$$

5.5 Time Complexity

Define Time Complexity $O_T[n]$ of Boolean Program $\mathcal{P}$

$$\mathcal{P} = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

5.6 Space Complexity

Define Space Complexity $O_S[n]$ of Boolean Program $\mathcal{P}$

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_S[n] := |\mathcal{M}| + |y_o| = M$$

5.7 Total Complexity as a Function of Time and Space Complexity

$$O[n] := |s^+[X_n]| = |\{\mathcal{L}, \mathcal{M}, y_o\}|$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o|$$

$$= O_T[n] + O_S[n]$$

5.8 $O_S[n] > 0^*$

Assuming Program is not void

5.8.1 Proof

Assume $O_S[n] = 0$

$$O_S[n] := |\mathcal{M}| + |y_o|$$

$$O_S[n] = 0 \Rightarrow \mathcal{M} = y_o = \varnothing$$

$$y_o = \varnothing; \;\; y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_S[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_S[n] \geqslant 0 \text{ by definition of magnitude}$$

$$\therefore O_S[n] > 0$$

5.9 $O_T[n] > 0^*$

Assuming Program is not void

5.9.1 Proof

Assume $O_T[n] = 0$

$$O_T[n] := |\mathcal{L}|$$

$$O_T[n] = 0 \Rightarrow y_o \notin \{\mathbb{T}, \mathbb{F}\}$$

$$y_o \notin \{\mathbb{T}, \mathbb{F}\};\ \ y_o \in \{\mathbb{T}, \mathbb{F}\} \text{ by definition of } s^+$$

$\therefore O_T[n] = 0$ contradicts the definition of solution s^+ of a decision problem

$$O_T[n] \geqslant 0 \text{ by definition of magnitude}$$

$$\therefore O_T[n] > 0$$

5.10 $O[n] > 0^*$

Assuming Program is not void

5.10.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0;\ \ O_S[n] > 0$$

$$\therefore O[n] > 0$$

5.11 $O[n] > O_T[n]^*$

Assuming Program is not void

5.11.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_S[n] > 0$$

$$\therefore O[n] > O_T[n]$$

5.12 $O[n] > O_S[n]^*$

Assuming Program is not void

5.12.1 Proof

$$O[n] := O_T[n] + O_S[n]$$

$$O_T[n] > 0$$

$$\therefore O[n] > O_S[n]$$

5.13 $O[n+1] \geqslant O[n]$

5.13.1 Proof

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$O[n] = |\, s^+[X_i]|$$
$$O[n+1] = \hat{O}[n] = |\, s^+[\hat{X}_i]|$$

For general solutions s^+

$$s^+[\hat{X}_i] \supseteq s^+[X_i]$$
$$\Rightarrow |s^+[\hat{X}_i]| \geqslant |s^+[X_i]|$$
$$\therefore \hat{O}[n] = O[n+1] \geqslant O[n]$$

6 Worst Case Complexity $O_{wc}[n]$

Let Program $\mathcal{P}$ with input space $X_n = \{x_1, x_2, ..., x_n\}$ and number of inputs n = $\tilde{n}$

$$O_{wc}[\tilde{n}] := O[\tilde{n}] : O[\tilde{n}] \geqslant O_i[\tilde{n}] \quad \forall X_{\tilde{n}} \in \mathbb{D}_{\mathcal{P}}$$

6.1 Proof of Existence

Let Program $\mathcal{P}$ with input space $\mathbb{X}_n$ having magnitude > 0
Prove the existence of $O_{wc}[n]$

7 Polynomial Complexity

7.1 Definition

Decision problem D with solution s^+ has polynomial total complexity $O[n]$ if

$$\exists K, C, \lambda_1 ... \lambda_K \quad :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

7.2 Polynomial Problems

Define $\mathbb{P}$, the set of Decision Problems that can be solved with Polynomial Complexity

$$\mathbb{P} := \{D_1, D_2, ...\} :$$

$$\exists K, C, \lambda_1 ... \lambda_K :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, D_i \in \mathbb{P}$$

7.3 Polynomial Order of Complexity

Solution s^+ with total complexity $O[n]$ is said to be of order n^K

$$O[n] \sim n^K$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

7.4 Property of Polynomial Complexity 1

Solutions with polynomial complexity have convergent complexity

$$lim_{n \to \infty} \frac{O[n+1]}{O[n]} = 1$$

7.4.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$
$$O[n+1] = (\lambda_K (n+1))^K + (\lambda_{K-1} (n+1))^{K-1} ... + \lambda_1 (n+1) + C$$
$$= (\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}$$

$$lim_{n \to \infty} \frac{O[n+1]}{O[n]}$$

$$= lim_{n \to \infty} \frac{(\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C}$$

$$= lim_{n\to\infty} \frac{(\lambda_K n)^K}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + \frac{(\tilde{\lambda}_{K-1} n)^{K-1}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + ... + \frac{\tilde{\lambda}_1 n}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C} + \frac{\tilde{C}}{(\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C}$$

$$= 1 = lim_{n\to\infty} \frac{O[n+1]}{O[n]}$$

7.5 Property of Polynomial Complexity 2

$$\exists K, \hat{C}, \hat{\lambda}_1, ..., \hat{\lambda}_{K-1} :$$
$$O[n+1] - O[n] = f_{n+1}[n] = (\hat{\lambda}_{K-1} n)^{K-1} ... + \hat{\lambda}_1 n + \hat{C} \quad \forall n$$

7.5.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$
$$O[n+1] = (\lambda_K (n+1))^K + (\lambda_{K-1} (n+1))^{K-1} ... + \lambda_1 (n+1) + C$$
$$= (\lambda_K n)^K + (\tilde{\lambda}_{K-1} n)^{K-1} ... + \tilde{\lambda_1} n + \tilde{C}$$
$$O[n+1] - O[n] = ((\tilde{\lambda}_{K-1} - \lambda_{K-1}) n)^{K-1} ... + (\tilde{\lambda_1} - \lambda_1) n + (\tilde{C} - C)$$
$$O[n+1] - O[n] = (\hat{\lambda}_{K-1} n)^{K-1} ... + \hat{\lambda}_1 n + \hat{C}$$

7.6 Property of Polynomial Complexity 3

$$limit_{n\to\infty} \frac{f_{n+1}[n]}{O[n]} = 0$$

7.6.1 Proof

$$limit_{n\to\infty} \frac{O[n+1]}{O[n]} = 1$$
$$limit_{n\to\infty} \frac{O[n] + f_{n+1}[n]}{O[n]} = 1$$
$$limit_{n\to\infty} \frac{O[n]}{O[n]} + \frac{f_{n+1}[n]]}{O[n]} = 1$$
$$limit_{n\to\infty} 1 + \frac{f_{n+1}[n]]}{O[n]} = 1$$
$$limit_{n\to\infty} \frac{f_{n+1}[n]]}{O[n]} = 0$$

7.7 Total Polynomial Complexity Implies Time bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

7.7.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$
$$O[n] := O_T[n] + O_S[n]; \;\; O_S[n] > 0$$
$$\therefore O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$

7.8 Total Polynomial Complexity Implies Space bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

7.8.1 Proof

$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$
$$O[n] := O_T[n] + O_S[n]; \;\; O_T[n] > 0$$
$$\therefore O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$

7.9 Polynomial Complexity in Space and Time Implies Polynomial Total Complexity

$$(O_S[n] == (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + \lambda_0)$$
$$\wedge$$
$$(O_T[n] == (\hat{\lambda}_M n)^M + (\hat{\lambda}_{M-1} n)^{M-1} ... + \hat{\lambda}_1 n + \hat{\lambda}_0)$$
$$\Rightarrow D \in \mathbb{P}$$

7.9.1 Proof

$$O_S[n] = \lambda_K n^K + \lambda_{K-1} n^{K-1} + ... + \lambda_1 n + \lambda_0$$
$$O_T[n] = \hat{\lambda}_M n^M + \hat{\lambda}_{M-1} n^{M-1} + ... + \hat{\lambda}_1 n + \hat{\lambda}_0$$
$$O[n] = O_S[n] + O_T[n]$$
$$^{*}O[n] = (\hat{\lambda}_0 + \lambda_0) + n(\lambda_1 + \hat{\lambda_1}) + ... + n^K(\lambda_K + \hat{\lambda_K}) + \hat{\lambda}_{K+1} n^{K+1} + ... + \hat{\lambda}_M n^M$$
$$\therefore O[n] \text{ has polynomial total complexity by definition}$$

* Assume K < M, similar proof for K=M, K>M

8 Non-Polynomial Complexity

8.1 Definition

Decision problem $\tilde{D}$ with solution s^+ has non-polynomial total complexity $O[n]$ if

$$\nexists K, \lambda_0, \lambda_1, ..., \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

8.2 Non-Polynomial Problems

Define $\mathcal{N}$, the set of Decision Problems that cannot be solved with Polynomial Complexity

$$\mathcal{N} := \{\tilde{D}_1, \tilde{D}_2, ...\} :$$
$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, \ s^+ \in S_i^+, \ \tilde{D}_i \in \mathcal{N}$$

8.3 $\mathbb{P}$ and $\mathcal{N}$ are disjoint

$$\mathbb{P} \cap \mathcal{N} = \varnothing$$

8.3.1 Proof

Let D $\in \mathcal{N}$

$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Assume D $\in \mathbb{P}$

$$\exists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Contradicts the definition of $\mathcal{N}$

$$\therefore D \in \mathcal{N} \Rightarrow D \notin \mathbb{P}$$

Let D $\in \mathbb{P}$

$$\exists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Assume D $\in \mathcal{N}$

$$\nexists K, C, \lambda_1 ... \lambda_K \ :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

Contradicts the definition of $\mathbb{P}$

$$\therefore D \in \mathbb{P} \Rightarrow D \notin \mathcal{N}$$

$$D \in \mathcal{N} \Rightarrow D \notin \mathbb{P}; D \in \mathbb{P} \Rightarrow D \notin \mathcal{N}$$
$$\therefore \mathbb{P} \cap \mathcal{N} = \varnothing$$

9 Divergent Complexity

See Appendix for additional proofs on Convergent and Divergent Functions

9.1 Divergent Programs

See Appendix for additional proofs on Convergent and Divergent Functions

9.2 Divergent Program $\Rightarrow$ Not-Polynomial Program

9.3 Not-Polynomial Program $\Rightarrow$ Divergent Programs

10 Solution to a Problem

10.1 Definition of a Solution

10.2 Complexity of a Solution

10.3 Worst Case Complexity of Solution s^+

10.4 Solution Space of a Question

10.5 Candidate Solutions

11 Optimal Complexity $O_{opt}[n]$

Let Problem $\mathcal{Q}$ with solution space $\mathbb{S}^+$ and number of inputs n = $\tilde{n}$

$$O_{opt}[\tilde{n}] := O[\tilde{n}] : O[\tilde{n}] \leqslant O_i[\tilde{n}] \quad \forall s_i^+[\tilde{n}] \in \mathbb{S}^+$$

11.1 Proof of Existence

Let Problem $\mathcal{Q}$ with solution space $\mathbb{S}^+$
Prove the existence of $O_{opt}[n]$

12 Polynomial Problems

13 Divergent Problems

AKA Not Polynomial Problems or Non-Polynomial Problems

14 Subfunctions

15 Subfunction Decomposition of Program

16 Subsolutions

16.1 Subfunction condition of solutions

Prove (xor state as a formal condition)

$$P[X_{n+1}] \supset P[X_n]$$

17 Inductive Complexity Expressions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1]$$

$$f_{n+1}[n] := O[n+1] - O[n]$$

17.1 Inductive Space and Time Formulas

$$f^T_{n+1}[n] := O_T[n+1] - O_T[n]$$
$$O_T[n+1] = O_T[n] + f^T_{n+1}[n]$$
$$f^S_{n+1}[n] := O_S[n+1] - O_S[n]$$
$$O_S[n+1] = O_S[n] + f^S_{n+1}[n]$$

17.2 Inductive Function Expressions

Relate $f_{n+1}[n]$ to equivalence functions

$$O[n] = O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1] = O[n] + f_{n+1}[n]$$
$$O_T[n] = O[n] - O_S[n]$$
$$O_S[n] = O[n] - O_T[n]$$

$$f_{n+1}[n] = O[n+1] - O[n]$$

$$f_{n+1}[n] = O_T[n+1] + O_S[n+1] - O[n]$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n]$$
$$f_{n+1}[n] = O[n+1] - O_T[n] - O_S[n]$$
$$f_{n+1}[n] = f^T_{n+1}[n] + f^S_{n+1}[n]$$

17.3 Zero Order Space Inductive Function

$$Let\ \ O_S[n] \sim n^0$$
$$f_{n+1}[n] = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n] = O_T[n+1] - O_T[n]$$

18 Discrete Derivative; Z Transform

18.1 Discrete Derivative

Define derivative for discrete function f[n]

$$\Delta_n^1 f[n] := f[n+1] - f[n]$$

We will use the above definition for the remainder of this document

18.2 Zero Order Derivative

$$\Delta_n^0 f[n] = f[n]$$

18.3 Kth Discrete Derivative

Define the Kth derivative of discrete function $f[n]$

$$\Delta_n^K f[n] := \Delta_n^{K-1} f[n+1] - \Delta_n^{K-1} f[n]$$

18.4 Kth Discrete Derivative as an Alternating Sum

$$\Delta_n^K f[n] := \Delta_n^{K-1} f[n+1] - \Delta_n^{K-1} f[n]$$

$$= (\Delta_n^{K-2} f[n+2] - \Delta_n^{K-2} f[n+1]) - (\Delta_n^{K-2} f[n+1] - \Delta_n^{K-2} f[n])$$

$$= (\Delta_n^{K-2} f[n+2] - 2\Delta_n^{K-2} f[n+1] - \Delta_n^{K-2} f[n])$$

$$= \textstyle\sum_{i=0}^{K} (-1)^j \ ({}_K C_j) \ \Delta_n^0 f[n+j]$$

$$= \textstyle\sum_{i=0}^{K} (-1)^j \ ({}_K C_j) \ f[n+j]$$

18.5 Z Transform

Define the Z Transform for discrete function f[n]

$$\mathcal{Z}(f[n]) := \textstyle\sum_{n=0}^{\infty} f[n] z^{-n}$$

18.6 Z Transform of 0 Order Derivative

$$\Delta_n^0 f[n] := f[n]$$

$$\mathcal{Z}(\Delta_n^0 f[n]) = \mathcal{Z}(f[n])$$

18.7 Z Transform of 1st Derivative

$$\Delta_n^1 f[n] := f[n+1] - f[n]$$

$$\mathcal{Z}(\Delta_n^1 f[n]) = \mathcal{Z}(f[n+1] - f[n])$$

$$= \sum_{n=0}^{\infty}(f[n+1] - f[n])z^{-n}$$

$$= \sum_{n=0}^{\infty}(f[n+1]z^{-n} - f[n]z^{-n})$$

$$= \sum_{n=0}^{\infty} f[n+1]z^{-n} - \sum_{n=0}^{\infty} f[n]z^{-n}$$

$$= \sum_{m=0}^{\infty} f[m+1]z^{-m} - \sum_{n=0}^{\infty} f[n]z^{-n}$$

Let

$$\hat{m} = m + 1; \;\; m = \hat{m} - 1$$

$$= \sum_{m=0}^{\infty} f[\hat{m}]z^{-(\hat{m}-1)} - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=1}^{\infty} f[\hat{m}]z^{-\hat{m}} - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=1}^{\infty} f[\hat{m}]z^{-\hat{m}} + f[0] - f[0] - \mathcal{Z}(f[n])$$

$$= z^1 \sum_{\hat{m}=0}^{\infty} f[\hat{m}]z^{-\hat{m}} - f[0] - \mathcal{Z}(f[n])$$

$$= z^1 \mathcal{Z}(f[n]) - f[0] - \mathcal{Z}(f[n])$$

$$\mathcal{Z}(\Delta_n^1 f[n]) = \mathcal{Z}(f[n])(z^1 - 1) - f[0]$$

18.8 Z Transform of Kth Derivative

$$\mathcal{Z}(f[n]) := \sum_{n=0}^{\infty} f[n]z^{-n}$$

$$\mathcal{Z}(\Delta_n^K f[n]) = \sum_{n=0}^{\infty} \Delta_n^K f[n]z^{-n}$$

$$= \sum_{n=0}^{\infty} \sum_{i=0}^{K} (-1)^j \; ({}_K C_j) \; f[n+j]z^{-n}$$

$$= \sum_{n=0}^{\infty}(f[n+K] - ({}_K C_1)f[n+K-1] + ({}_K C_2)f[n+K-2] - ... \pm f[n])z^{-n}$$

$$= \sum_{n=0}^{\infty} f[n+K]z^{-n} - ({}_K C_1)f[n+K-1]z^{-n} + ({}_K C_2)f[n+K-2]z^{-n} - \\ ... \pm f[n]z^{-n}$$

$$= z^K \mathcal{Z}(f[n]) + \sum_{i=0}^{K-1} f[i] - ({}_K C_1)z^{K-1}\mathcal{Z}(f[n]) - \sum_{j=0}^{K-2} f[j] + \\ ({}_K C_2)z^{K-2}\mathcal{Z}(f[n]) + \sum_{k=0}^{K-3} f[k] - ... \pm \mathcal{Z}(f[n])$$

When K is odd

$$\mathcal{Z}(\Delta_n^K f[n]) = (z-1)^K \mathcal{Z}(f[n]) + \sum_{i=0}^{\frac{n+1}{2}} f[2i] \quad K > 0$$

When K is even

$$\mathcal{Z}(\Delta_n^K f[n]) = (z-1)^K \mathcal{Z}(f[n]) + \sum_{j=0}^{\frac{n}{2}} f[2j+1] \quad K > 0$$

Computation Basis

19 Definition of Basis

20 Definition of Necessary Instructions

$$s_i \in S_j^+ \quad \forall S_j^+ \in \mathbb{S}_Q$$

21 Definition of Convertible Instructions

$$s_i \in S_j^+ \quad "for\ some"\ S_j^+ \in \mathbb{S}_Q$$

22 Theorem of Polynomial Subfunctions

23 Theorem of Divergent Subfunctions

24 Two integer Sum to N

24.1 State formal definition of Sum to N : $x_i + x_j == N$

$$X_n = \{x_1, ..., x_n\}$$

$$D := f[X_i, N] = a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_n] = P[X_n] :$$
$$(P[X_i] = y_o == a_o \quad \forall X_i) \quad \cap \quad (P[X_{n+1}] \supseteq P[X_n] \quad \forall X_{n+1})$$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$D = f[X_i] = \quad \exists x_j, x_k \in X_n \quad j \neq k :$$
$$x_j + x_k == N$$

24.2 Express a formal solution : $O_S[n] \sim n^0$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i < n \ , \ n \geqslant j > i$$
$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$
$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$

24.3 Prove s^+ satisfies the subfunction condition of solutions: $P[X_{n+1}] \supseteq P[X_n] \quad \forall X_{n+1}$

$$X_n = \{x_1, x_2, ..., x_n\}; \quad X_{n+1} = \{x_1, x_2, ..., x_n, x_{n+1}\}$$
$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s^+_{n+1} = s^+ \cup \hat{s}^+$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i < n \ , \ n \geqslant j > i$$

$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$

$$\forall k < n+1$$
$$s... = b_1 \leftarrow x_k + x_{n+1}$$
$$s... = b_1 \leftarrow b_1 == N$$
$$s... = y_o \leftarrow y_o \vee b_1$$

$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$
$$\hat{s}^+ = \{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$
$$s^+_{n+1} = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\} \ \cup$$
$$\{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$

$$s^+_{n+1} = s^+ \cup \hat{s}^+ = P[X_{n+1}] \supseteq P[X_n] = s^+$$

24.4 Determine $O[n], O_S[n], O_T[n], f_{n+1}[n], f^T_{n+1}[n], f^S_{n+1}[n]$ for the above solution

$$O_S[n] = |y_o| + |b_1| = 2$$
$$O_T[n] = 3n(n-1) + 1 = 3n(n-1) - 1 + O_S[n]$$
$$O[n] = 3n(n-1) + 3 = 3n^2 - 3n + 3$$

$$f^S_{n+1}[n] = 0$$
$$f^T_{n+1}[n] = 6n$$
$$f_{n+1}[n] = f^S_{n+1}[n] + f^T_{n+1}[n]$$

24.5 Verify $O[n+1] = O[n] + f_{n+1}[n]$

$$O[n+1] = O[n] + \hat{O}[n]$$
$$3(n+1)^2 - 3(n+1) + 3 = 3n^2 - 3n + 3 + 6n$$
$$3n^2 + 6n + 3 - 3n - 3 + 3 = 3n^2 + 3n + 3$$

$$3n^2 + 3n + 3 = 3n^2 + 3n + 3$$

24.6 Show s^+ has Polynomial Complexity by the definition of Total Polynomial Complexity

$$O[n] = 3n^2 - 3n + 3$$

24.7 Show the limit $_{n\to\infty}\frac{O[n+1]}{O[n]}$ does not Diverge

$$limit_{n\to\infty}\frac{O[n+1]}{O[n]} =$$

$$limit_{n\to\infty}\frac{3n^2+3n+3}{3n^2-3n+3} =$$

$$limit_{n\to\infty}(\frac{3n^2-3n+3}{3n^2-3n+3} + \frac{6n}{3n^2-3n+3}) =$$

$$limit_{n\to\infty}(1 + \frac{6n}{3n^2-3n+3}) = 1$$

25 The Knapsack Problem

25.1 The Knapsack Problem

The Knapsack Problem is a famous problem in computer science which asks if objects can be stored in a knapsack. Typically the problem is designed with two constraints, weight and value. Given objects x_i, each with a respective weight w_i and value v_i, does there exist a combination of objects lighter than input weight W and more valuable than input value V?

25.2 Formal Definition

$$X_n = \{x_1, x_2, ..., x_n\} = \{\{w_1, v_1\}, \{w_2, v_2\}, ..., \{w_n, v_n\}\}$$

$$I = \{i_1, i_2, ..., i_n\} : i_l \in \{0, 1\} \ \ \forall i_l \in I$$

$$D := f[X_n, W, V] = a_o \in \{\mathbb{T}, \mathbb{F}\} = \exists I :$$
$$(\textstyle\sum_{j=1}^{n} i_j w_j < W) \wedge (\sum_{j=1}^{n} i_j v_j \geqslant V)$$

25.3 Solve for C[n]

25.3.1 Expressing I as a binary number

$$I = \{i_1, i_2, ..., i_n\} : i_l \in \{0, 1\} \ \ \forall i_l \in I$$

Valid combinations of I

$$I_{valid} =$$
$$\{\{0, 0, 0, ..., 0, 0, 1\}, \{0, 0, 0, ..., 0, 1, 0\}, \{0, 0, 0, ..., 0, 1, 1\}, ..., \{1, 1, 1, ..., 1, 1, 1\}\}$$

$$C[n] = |I_{valid}[n]| = 2^n - 1$$

25.3.2 Using a sum of combinations of inputs x_i

$$X_n = \{x_1, x_2, ..., x_n\} = \{\{w_1, v_1\}, \{w_2, v_2\}, ..., \{w_n, v_n\}\}$$

Valid combinations of x_i

$$X_{valid}[n] =$$

$$\{x_1\}\cup\{x_2\}\cup...\cup\{x_n\}\cup\{x_1,x_2\}\cup\{x_1,x_3\}\cup...\cup\{x_{n-1},x_n\}\cup...\cup\{x_1,x_2,...,x_n\}$$

$$= \ {}_{X_n}C_1 \cup \ {}_{X_n}C_2 \cup ... \cup \ {}_{X_n}C_n$$

$$C[n] = |X_{valid}[n]| = \sum_{j=1}^{n} \ {}_nC_j$$

25.3.3 Verify consistency

$$C[n] = |X_{valid}[n]| = |I_{valid}[n]|$$

$$= 2^n - 1 = \sum_{j=1}^{n} \ {}_nC_j = {}_nC_1 + {}_nC_2 + ... + {}_nC_n$$

$$= 2^n - 1 = 2^n - 1$$

25.4 Express a solution s^+ of the Knapsack Problem

25.5 Prove s^+ satisfies the subfunction condition of solutions

25.6 Determine $O[n], O_T[n], O_S[n], f_{n+1}[n]$

25.7 Show $s^+ \notin \mathbb{P}$

25.8 Express the Solution Space $\mathbb{S}$ for The Knapsack Problem

25.9 Prove a lower bound for all solutions $s^+ \in S^+ := O_{lower}[n]$

25.10 Prove D has Divergent Complexity

Parallel Programs

26 Workers

27 Definition of a Parallel Program

Optimal Programs

28 Optimal Complexity

29 Optimal Solutions

Appendix

30 Domains

31 Bounded Input Bounded Output

In abstraction theory, functions have a domain (bounded set of every possible input and output)

Convergence, Divergence, Divergent Complexity

32 Convergent Functions

33 Divergent Functions

References

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